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GitHub Link	https://github.com/kiduuuu/MLA0405-Deep-learning/tree/main

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA04**
SLOT :

COURSE OUTCOME COVERED

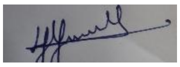
CO3: Analyse and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I **D.Chinna Moulali** *<u>/ 192425278</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: **Name of the student:** *S. Yuva Guru Mani Varma*

Criteria	**Conceptual Understanding (2.5)**	Linkages **(2.5)**	Organization **(2)**	Integration of Literature **(2)**	**Clarity & Presentation *(1)***	**Total (10)**
Weightage	***25 %***	***25%***	***20%***	***20 %***	***10 %***	**100%**
Q1						
Q2						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS: 02 **Total Marks: 20**

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

1. Suitable CNN Architecture

A suitable CNN for chest X-ray classification can be designed as follows:

Input Image (224 × 224 × 1)
→ **Convolution Layer 1:** 32 filters, 3×3, ReLU
→ **Batch Normalization**
→ **Max Pooling (2×2)**
→ **Convolution Layer 2:** 64 filters, 3×3, ReLU
→ **Batch Normalization**
→ **Max Pooling (2×2)**
→ **Convolution Layer 3:** 128 filters, 3×3, ReLU
→ **Max Pooling (2×2)**
→ **Flatten**
→ **Fully Connected Layer: 128 neurons, ReLU**
→ **Dropout**
→ **Output Layer: 2 neurons with Softmax**

The two output classes are **Normal** and **Disease-Affected**.

2. Motivation and Role of CNN Components

CNNs are suitable because they automatically learn visual features from X-ray images without requiring manual feature extraction.

- **Convolutional layers:** Apply filters to detect features such as edges, textures, shapes and abnormal regions.
- **Filters/Kernels:** Small matrices that scan the image and generate feature maps. Different filters learn different visual patterns.
- **ReLU activation:** Introduces non-linearity and helps the network learn complex patterns.
- **Pooling layers:** Reduce the spatial dimensions of feature maps, decreasing computation while retaining important features. Max pooling is commonly used.
- **Fully connected layers:** Combine the learned features and perform the final classification.
- **Softmax output:** Produces probabilities for the two classes.

Early layers learn simple features such as edges, while deeper layers learn more complex patterns associated with possible diseases.

3. Parameter Sharing

In a fully connected network, every neuron is connected to every pixel, resulting in a very large number of parameters. CNNs use the same filter weights at different image locations.

For example, a **3×3 filter has only 9 weights plus a bias**, regardless of the image size. The same filter is reused across the entire image.

Therefore, parameter sharing:

- Greatly reduces trainable parameters.
- Reduces memory and computational requirements.
- Allows the same feature to be detected at different locations.
- Improves the network's ability to recognize spatial patterns.

4. Regularization Techniques

Medical image datasets can contain variations and may cause overfitting. The following techniques are useful:

Data Augmentation:

Images can be randomly rotated, shifted, cropped, scaled or slightly adjusted in brightness. This creates more varied training examples and improves generalization.

Dropout:

Randomly disables a percentage of neurons during training. This prevents the network from depending too heavily on particular neurons.

Batch Normalization:

Normalizes intermediate activations during training. It improves training stability and can provide a regularization effect.

Other methods such as **early stopping, weight decay and transfer learning** can also reduce overfitting.

5. AlexNet vs ResNet

Feature	**AlexNet**	**ResNet**
Depth	8 learned layers	Can have 18, 34, 50, 101+ layers
Feature learning	Good	Excellent
Training	Relatively simple	More complex but easier for deep networks
Vanishing gradient	More susceptible	Reduced using residual connections
Computational cost	Lower	Higher depending on version
Classification performance	Good	Generally better for complex images

ResNet is more appropriate for medical X-ray classification. Its deeper architecture can learn more detailed and hierarchical features, while **residual/skip connections** help solve the vanishing-gradient problem and make deep networks easier to train.

A pretrained ResNet can also be fine-tuned on chest X-ray data, reducing training time and improving performance when the medical dataset is not extremely large. However, if computational resources are very limited, a smaller ResNet such as **ResNet-18** can be selected.

Conclusion: ResNet is preferred because medical X-rays contain complex visual patterns, and deeper residual networks generally provide stronger feature learning and classification performance.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)
An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

1. CNN Architecture for Autonomous Vehicles
A CNN-based object recognition system can use the following structure:
Camera Image Input
→ Convolution + ReLU
→ Pooling
→ Convolution + ReLU
→ Pooling
→ Deeper Convolution Layers
→ Feature Extraction
→ Fully Connected/Detection Head
→ Object Classes and Bounding Boxes
The system can be trained to identify **cars, pedestrians, bicycles, traffic signs and road obstacles**.
For a real autonomous vehicle, a modern object-detection model based on a ResNet backbone would generally be more suitable than using a simple image-classification CNN alone because the vehicle needs to determine **what the object is and where it is located**.

2. Purpose of CNN Layers
Input Layer:
Receives camera images, usually resized to a standard resolution and normalized.
Convolutional Layers:
Apply filters to detect important visual patterns.
Activation Layer:
ReLU introduces non-linearity, allowing the network to learn complex relationships.
Pooling Layers:
Reduce feature-map size and computational cost while retaining important information.

Fully Connected/Output Layers:
Use extracted features to classify objects. In an object detector, the detection head also predicts object locations.
Feature learning occurs progressively:

- **Early layers:** Edges, corners and simple textures.
- **Middle layers:** Shapes, patterns and object parts.
- **Deep layers:** Complete objects such as cars, pedestrians and traffic signs.

3. Importance of Parameter Sharing
CNN filters are reused across different image locations. Therefore, a filter that learns to detect a particular edge or pattern can detect that feature anywhere in the image.
Parameter sharing:

- Reduces the number of trainable parameters.
- Reduces memory and computational requirements.
- Makes CNNs more efficient than fully connected networks.
- Provides translation-related feature detection.
- Allows the same object features to be recognized at different image positions.

This is especially important for autonomous vehicles because objects can appear anywhere in the camera frame.
4. Overfitting and Regularization
The vehicle dataset may contain thousands of images but still have limited examples of certain weather, lighting or road conditions. Overfitting can therefore occur.
Suitable methods include:

- **Data augmentation:** Changes brightness, contrast, scale, rotation and cropping to simulate different conditions.
- **Dropout:** Randomly removes neurons during training.
- **Batch normalization:** Stabilizes training and can reduce overfitting.
- **Weight decay:** Penalizes excessively large model weights.
- **Early stopping:** Stops training when validation performance stops improving.
- **Transfer learning:** Starts with a CNN pretrained on a large dataset and fine-tunes it for road-object recognition.

These methods improve generalization to new road environments.
5. AlexNet vs ResNet

Feature	**AlexNet**	**ResNet**
Architecture	Relatively shallow CNN	Deep residual CNN
Depth	8 learned layers	18, 34, 50, 101+ layers
Parameters	Relatively large for its depth	More efficient feature learning
Training difficulty	Simpler	Residual connections make deep training easier
Vanishing gradients	More susceptible	Skip connections reduce the problem
Feature learning	Good basic features	Strong hierarchical features
Accuracy	Lower for complex modern tasks	Generally higher
Computational cost	Lower	Depends on ResNet version

AlexNet was historically important because it demonstrated the effectiveness of deep CNNs, but it is relatively shallow and less suitable for modern autonomous-vehicle perception.

ResNet uses **residual/skip connections**, allowing information and gradients to flow through deeper networks. This makes it possible to learn more complex features without suffering as severely from vanishing gradients.

6. Recommended Architecture

ResNet is the better choice for an autonomous vehicle system, particularly as a backbone for an object-detection network.

The main reasons are:

1. **Higher accuracy:** Deep residual networks can learn detailed features needed to distinguish similar objects.
2. **Better feature learning:** They learn hierarchical features from simple edges to complex objects.
3. **Reduced vanishing-gradient problem:** Skip connections make deep networks easier to train.
4. **Transfer learning:** Pretrained ResNet models can be fine-tuned using road-scene datasets.
5. **Real-time possibility:** A smaller model such as ResNet-18 or ResNet-34 can provide a useful balance between accuracy and computational cost.

However, autonomous vehicles require **real-time processing**, so the largest ResNet models may be too computationally expensive. A lightweight ResNet-based detector or optimized detection architecture should therefore be selected according to the vehicle's GPU/AI hardware and required frame rate.

Conclusion: ResNet provides a better balance of accuracy, feature-learning capability and training stability. For real-time autonomous driving, a **lightweight ResNet backbone with an object-detection head** would be more appropriate than directly using AlexNet.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

		reasoning and insights			
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand